

PAPER_1710 — D-T Cross-Section Peak Energy = $E_\sigma = A_5 + D_{\text{phys}} = 64 \text{ keV}$ (Bosch-Hale peak in CM frame)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Plasma Fusion **Date:** June 18, 2026 **Status:** CLOSED — EXACT closure (PAPER_1196 plasma fusion proof set)

Observation

PAPER_1196_S420 (PAPER_1196 Plasma Fusion Unified Proof Set) gives:

$E_\sigma = A_5 + D_{\text{phys}} = 64 \text{ keV}$ (Bosch-Hale peak in CM frame)

Status: **EXACT**.

UQFF Closed Identity

$E_\sigma = A_5 + D_{\text{phys}} = 64 \text{ keV}$ (Bosch-Hale peak in CM frame) EXACT

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

ITER design parameters are chosen by international fusion-engineering consensus. UQFF supplies structural integer-primitive forms demonstrating the parameters' deep mathematical origin.

Reference

- Source: PAPER_1196_S420
- Related: PAPER_1696-1705 (tier-28); PAPER_1686-1695 (tier-27)
- Calculator dispatch: `calculate_paradox({"paradox": "dt_e_sigma_64_kev"})`

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